Technology stack

Clear separation of interface, application logic, data and deployment.

01 / TECHNOLOGY

Category	Used in the system	Purpose
UI / design	CSS, Lucide React, Framer Motion	Responsive layouts, icons, transitions and visual feedback.
Frontend	React 19, React Router 7, Vite 7	Components, forms, cart state and role pages; Vite builds the web app.
Backend	Supabase Edge Functions (Deno); PostgreSQL functions / RPC	Server-side validation, role checks, order actions and business rules. Optional Node.js / Express service supports server OCR.
Database	PostgreSQL through Supabase	Stores users, orders, payments, menu, inventory, suppliers and audit records.
Data backup	GitHub Actions → encrypted archives → Google Drive. Daily: 2:00 AM Philippine time; retention: 90 days.	Full database daily; new / changed storage files daily; full storage monthly and on manual runs. Checksums verify files. Restore uses the recovery key and required snapshots / increments; admin monitors status.
File storage / sync	Supabase Storage + Realtime	Stores images and documents; pushes order and stock changes to connected screens.
Languages	JavaScript / JSX, TypeScript, SQL, HTML, CSS	JS/JSX: React app; TypeScript: Edge Functions; SQL: schema / rules; HTML/CSS: structure and styling.
Security	Supabase Auth; RLS; role checks; signed file URLs; MFA*	Authentication, record ownership, restricted actions and private-file access. Validate inputs and keep secrets on the backend.
Version control / collaboration	Git + GitHub	Track code changes, share the repository and coordinate development through commits and branches.
Domain	Hostinger - thecoffeerealm.store	Domain provider specified for this reviewer.
Web hosting	Hostinger - frontend deployment target	Serves the built website. Supabase separately provides backend, database and file-storage services.

*Admin MFA and portal-session enforcement require their configured rollout. Hostinger is the requested deployment setup; live hosting was not verified.

APIs & special functions

Which service is used, what it does, and what the result means.

02 / INTEGRATIONS

Integration	API / technology	Function
Core services	Supabase Auth, Data API / RPC, Storage and Realtime	Sign-in, authorized data access, server business actions, uploads and live updates.
Google sign-in	Google OAuth via Supabase Auth	Customer authentication using a Google account.
Map / address	Leaflet + OpenStreetMap; Nominatim API; browser geolocation	Display maps, search / reverse-geocode addresses and select a delivery pin within supported areas.
Order emails	Gmail SMTP + Nodemailer	Send order notifications and customer-message replies. SMTP is an email protocol, not a REST API.
Raimu assistant	support-chat Edge Function; Groq, Cloudflare Workers AI, Gemini fallback	Internal role-aware help, navigation and report assistance; provider depends on configured keys.

FX Special functions

Key mechanisms to explain during the defense.

OCR scanning — Tesseract.js reads a payment image in the browser. Canvas improves contrast; pattern matching extracts reference digits. The user checks or corrects the result.

Receipt generation — Order data becomes an itemized HTML receipt: items, quantities, tax, discounts, fees, total and payment details. Browser Print provides printing / Save as PDF; no receipt API is needed.

Report exports — jsPDF generates report PDFs; ExcelJS creates spreadsheets. Filters determine the reporting period and records included.

Stock / loyalty — Recipe and product links connect sales to stock usage. Realm Passport links customer purchases, displays progress and records eligible reward claims through backend RPCs.

> **Payment boundary:** GCash / bank-transfer screenshots are reviewed manually; OCR does not prove payment. COD is available for eligible delivery orders. Recording a refund does not itself transfer money.

Customer experience

From discovering the café to ordering, tracking and earning loyalty rewards.

03 / CUSTOMER SCOPE

01 Storefront & ordering

Guests can explore public pages; checkout and personal records require a customer account.

Landing page	Explore the hero and Realm Tour link, featured menu, café story / timeline, amenities, ordering guide, testimonials, social posts, location and store contacts.
Account & profile	Register / verify an account or use Google sign-in. Manage profile, photo, password and saved / default addresses. Account deletion checks eligibility.
Menu & cart	Search / filter products; select variants, temperature, sugar, ice and add-ons. Add instructions, adjust quantities and review prices and stock availability.
Checkout	Choose pickup or delivery and a schedule. Search / pin an address, check the supported area, delivery fee, benefit discount and final total before placing the order.
Payment & tracking	Upload GCash / bank proof; scan and confirm reference digits, or choose eligible COD. Follow order status / tracking links, confirm delivery received and view receipts.
Order history	Cancel eligible early orders; payment-related cases enter review. View cancellation / refund status, reorder available items and submit feedback for completed orders.
Benefits & support	Apply for Senior / PWD benefits with details and documents; track decisions / resubmissions. Read help, policies and café information, or send a contact inquiry.

RP Realm Passport

Digital loyalty passport for the account owner; usable online and at the counter.

Identity & stamps	Open the page-turn passport with Realm ID, member details, purchase stamps and reward progress. The stated rule is one stamp per completed, paid order.
Rewards & counter	Every five stamps unlocks the next reward. Show the Realm ID at checkout to link a counter sale; cashier checks eligibility and records claims.
Download	Save the passport with current stamps as a PowerPoint (.pptx) copy. The live account and backend claim history remain the reference for redemption.

Cashier workspace

Walk-in sales, accurate payment recording, itemized receipts and loyalty service.

04 / CASHIER SCOPE

02 Point of sale

Access: cashier POS and its transaction / passport tools.

Order setup	Start dine-in or takeout transactions and manage multiple order tabs. Keep each customer cart separate while switching between orders.
Product selection	Browse / search available menu items and select quantities, variants, temperature, sugar, ice and add-ons. Review customizations before adding to the order.
Cart & totals	Edit quantities or remove items. Review subtotal, discount, VAT / exempt amounts and final total. Check availability before saving the transaction.
Cash payment	Enter cash received; require enough to cover the total and calculate change. Store tendered amount and change with the payment record.
Digital payment	Record GCash or bank transfer and required references / bank details. Reference validation supports recording; it is not bank settlement verification.
Senior / PWD	Record the benefit type, customer name and ID. Select eligible order lines; apply the configured discount and VAT-exempt calculation to those items.
Realm Passport	Find a member by Realm ID and link the walk-in sale. View purchase / claim history; check backend reward eligibility and record an eligible claim.
Receipts & history	Save the sale and generate a receipt with item details, totals, payment and change. Print / Save as PDF; review recent transactions and reopen receipts.
Live updates & help	Receive menu / order / payment refreshes through Realtime. Use Raimu for role-aware guidance, payment issues and recent-transaction questions.

> **Sale flow:** Build cart → apply eligible benefit → record payment → save sale → issue receipt. Link the Realm ID when the customer wants the counter purchase credited.

> **Role boundary:** The cashier handles sales and reward claims. Employee access, store settings and management approvals belong to the administrator.

Operations staff

Daily fulfillment, stock management, purchasing, menu maintenance and records.

05 / STAFF SCOPE

03 Operational modules

Access: Order Preparation, Inventory, Purchase Orders, Manage Menu, Transactions and Settings.

Order queue	Filter active, completed and cancelled orders. Inspect items, customizations, instructions, customer details, fulfillment method and scheduled time.
Payment review	Open manual payment proof through restricted file access; compare references and order details before confirming the order. OCR does not approve payment.
Preparation	Advance accepted orders through preparation, ready-for-pickup / delivery and completion. Save a delivery tracking link; follow the permitted status transitions.
Cancel / refund	Review eligibility, reason and payment outcome. Approve / reject cancellation; record refund completion with a reference and resolve the case. Normal cancellation stops at preparation.
Inventory	Maintain ingredients and finished products, quantities, units and stock thresholds. Record stock adjustments with movement type / reason; inspect movement history and recipe usage.
Stock planning	Configure recipe quantities and linked products / variants. Maintain daily opening-stock targets and timing; monitor low / out-of-stock items and related alerts.
Purchase orders	Manage suppliers; draft items, quantities, estimated costs, delivery date and notes. Submit for admin approval; revise rejected requests and mark approved orders sent.
Receiving	Check delivery conditions and attach proof, invoice and supplier receipt. Receive accepted deliveries into stock; report / return problem deliveries without adding rejected stock.
Manage menu	Maintain categories, products, prices, images, variants, add-ons, recipes, availability and benefit eligibility. Submit approval-controlled changes for admin review.
Transactions	Search / filter order and payment records; inspect details, cancellation / refund information and receipts. Print or download receipts for operational reference.
Settings & help	Edit personal profile / password and notification, theme, motion or contrast preferences. Review sessions, receive order / stock alerts and use Raimu for guidance.

Administrator

Business oversight, approval authority, reporting, security and configuration.

06 / ADMIN SCOPE

04 Administration modules

Admin reviews and governs operations; staff performs day-to-day order handling and receiving.

Dashboard	Monitor sales, fulfillment mix, recent orders, bestselling items, low stock and pending issues; open the corresponding management module.
Inventory & reports	Inspect ingredient / finished-product quantities, reorder levels, low / out-of-stock status and movements. Filter and export inventory reports for stock activity and usage.
Purchase orders	Review requested items, costs and delivery dates; approve / reject with feedback. Mark approved orders sent and inspect delivery documents, issues and activity.
Menu approvals	Review pending additions, edits, removals and discount-eligibility requests. Approve or reject proposed changes before controlled updates take effect.
Benefits verification	Inspect Senior / PWD applications and documents. Approve, reject or request resubmission with a reason; manage approved benefit records.
Transactions / refunds	Search transaction history and inspect payments / receipts. Filter and export the cancellation / refund report to review reasons, refund status and financial impact.
Sales & analytics	Filter periods; compare sales, products, categories, payment methods, order channels and buying times. Export reports. The Analytics route shares the sales-report module.
Content & messages	Manage homepage hero, featured menu, café story, testimonials, footer and social links. Preview the storefront; review customer inquiries and send email replies.
Users & audit	Create / edit / remove employees, assign roles and send resets. Search and export audit activity to trace actions and account changes.
System settings	Set store contacts, operating hours, ordering / fulfillment availability, delivery zones / fees, payment instructions / QR assets, COD limit and pricing / VAT policy.
Backups & security	Inspect Google Drive backup history, status, database size and file counts. Manage MFA / recovery codes and team-session revocation when enabled; adjust personal preferences.

> **Control flow:** Staff submits → admin reviews → approved action proceeds. Database permissions enforce access; a hidden button alone is not security.